Threat Landscape Types of cyber threats: Phishing, ransomware, insider threats, etc. Practical: Simulate a phishing attack in a controlled environment using open-source tools (e.g., SET toolkit). Demonstrate email spoofing and prevention techniques.

Tasks

1. Create a table listing 3 real-world examples each of phishing, ransomware, and insider threats that could affect apps like WhatsApp, Flipkart, or Paytm. For each, describe in 1-2 lines how the attack works.

Ans.:-

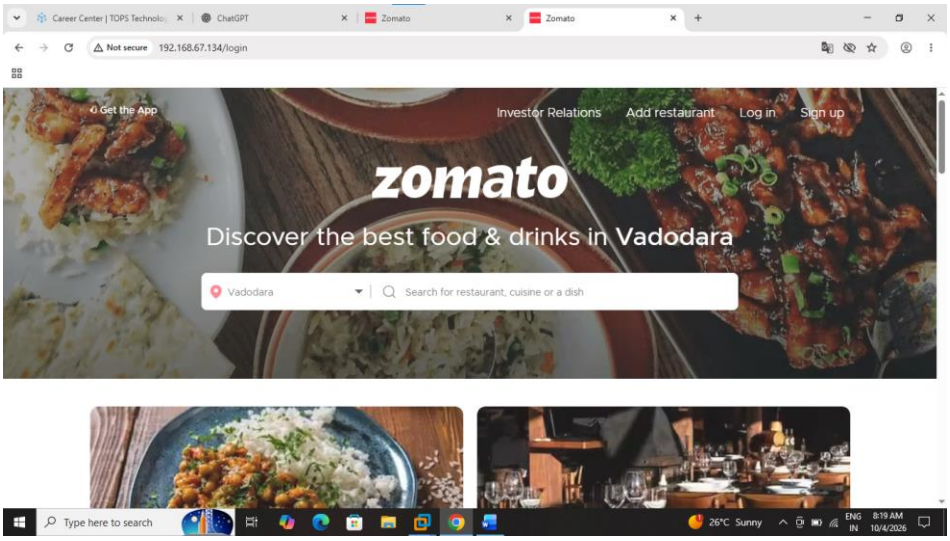
Threat	Real-world example	How the attack works
Phishing	Fake WhatsApp verification message	An attacker sends a fake message claiming to be WhatsApp support and asks for the OTP or verification code. If the victim shares it, the attacker may take over the account.
Phishing	Fake Flipkart offer/link	A victim receives a fake “special discount” link that looks like Flipkart. The fake website collects login, card, or UPI details.
Phishing	Fake Paytm KYC message	An attacker sends an SMS/email saying the Paytm account will be blocked unless KYC is completed. The link leads to a fake page designed to steal credentials or payment information.
Ransomware	Ransomware on a company's WhatsApp-support systems	Malware encrypts important files or systems used by a company. Attackers demand payment to provide a decryption key.
Ransomware	Ransomware affecting an e-commerce company's servers	A malicious attachment or compromised account gives attackers access to internal systems, where they encrypt databases or servers and demand a ransom.

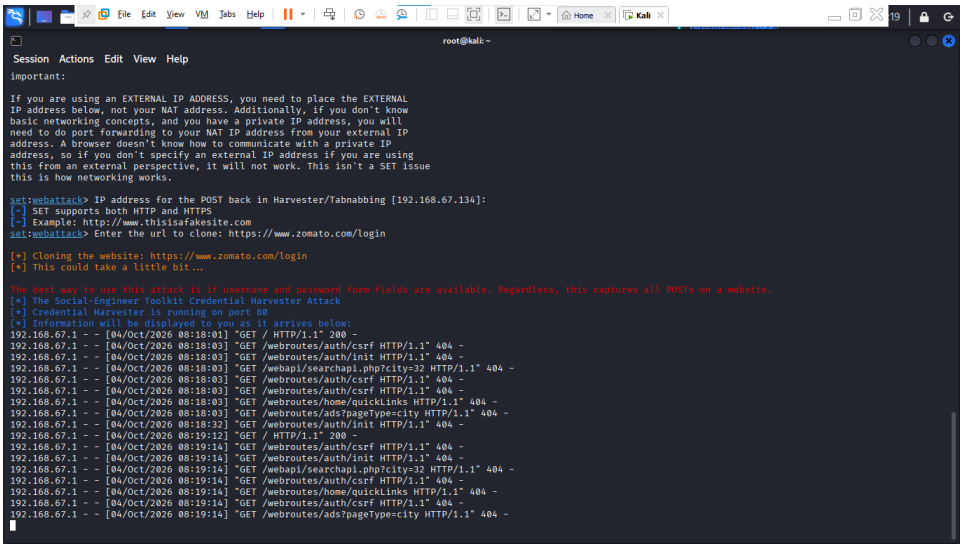
Threat	Real-world example	How the attack works
Ransomware	Ransomware affecting a digital-payment company's infrastructure	Attackers exploit a vulnerable system and encrypt critical infrastructure, potentially disrupting payment-related services until systems are restored.
Insider Threat	Employee misuses WhatsApp customer information	An employee with legitimate access intentionally copies or shares customer information with an unauthorized person.
Insider Threat	Flipkart employee leaks customer/order data	An employee abuses authorized database access and exports customer or order information for unauthorized use.
Insider Threat	Paytm employee misuses access credentials	An employee uses legitimate access to view, copy, or disclose sensitive customer or transaction information without authorization.

**2. Use the Social Engineering Toolkit (SET) to simulate a basic phishing page that mimics a login screen for a popular app (e.g., Instagram or Zomato). Capture and display the entered credentials in your local environment.

Hint: Use the 'Website Attack Vectors' option in SET and select 'Credential Harvester Attack Method'.**

Ans.:-

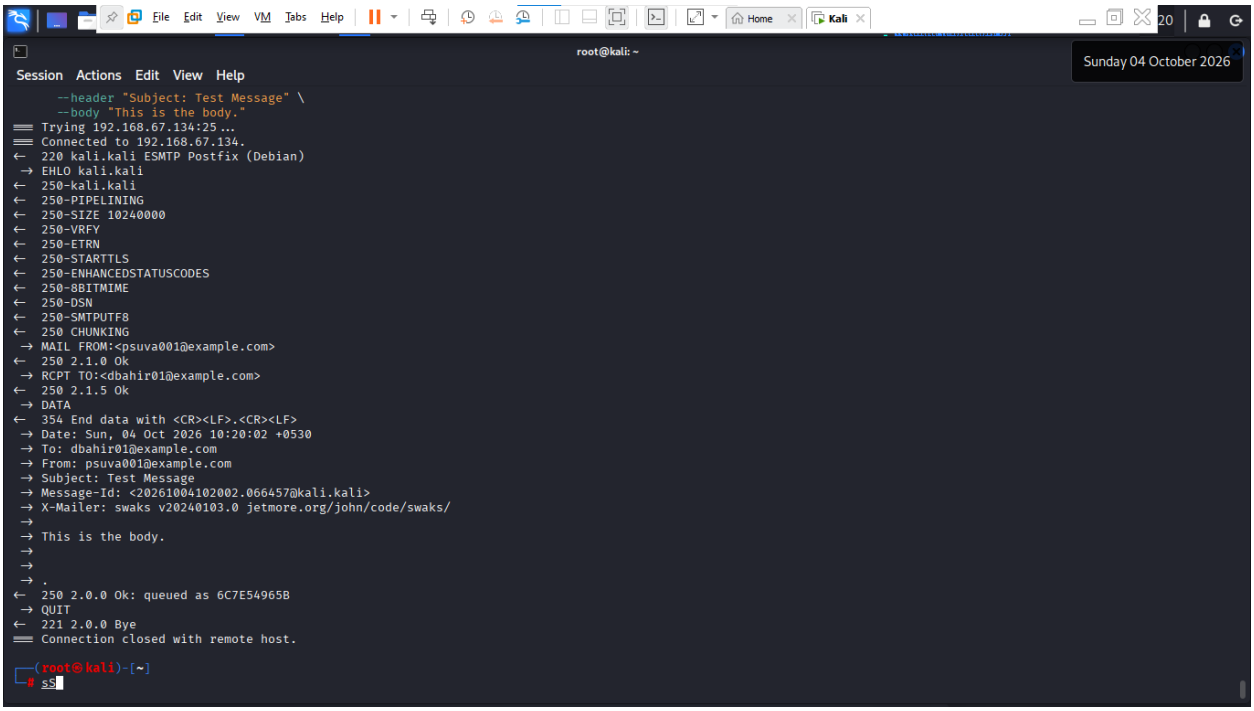




**3. Demonstrate email spoofing by sending a test email to yourself using a tool like 'sendmail' or an online SMTP testing service. Show how the sender's name can be faked to look like a message from BookMyShow or IRCTC.

Constraint: Do not use your real credentials or send emails to others — only test on your own email in a controlled setup.**

Ans.:-

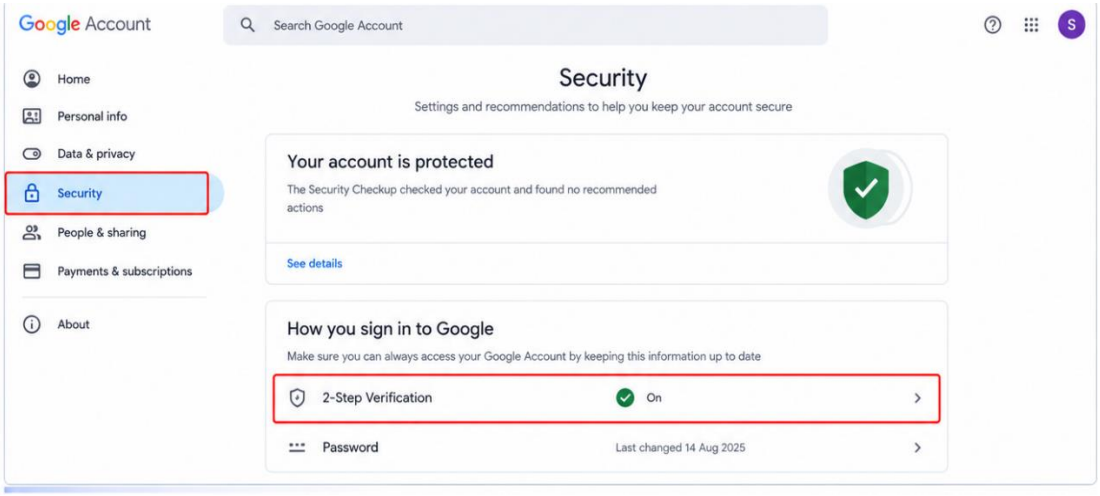


4. Research and implement at least two prevention techniques against phishing or email spoofing (such as enabling 2FA or checking email headers) on your own email account. Take screenshots or write a short note explaining how you applied each technique.

Ans.:-

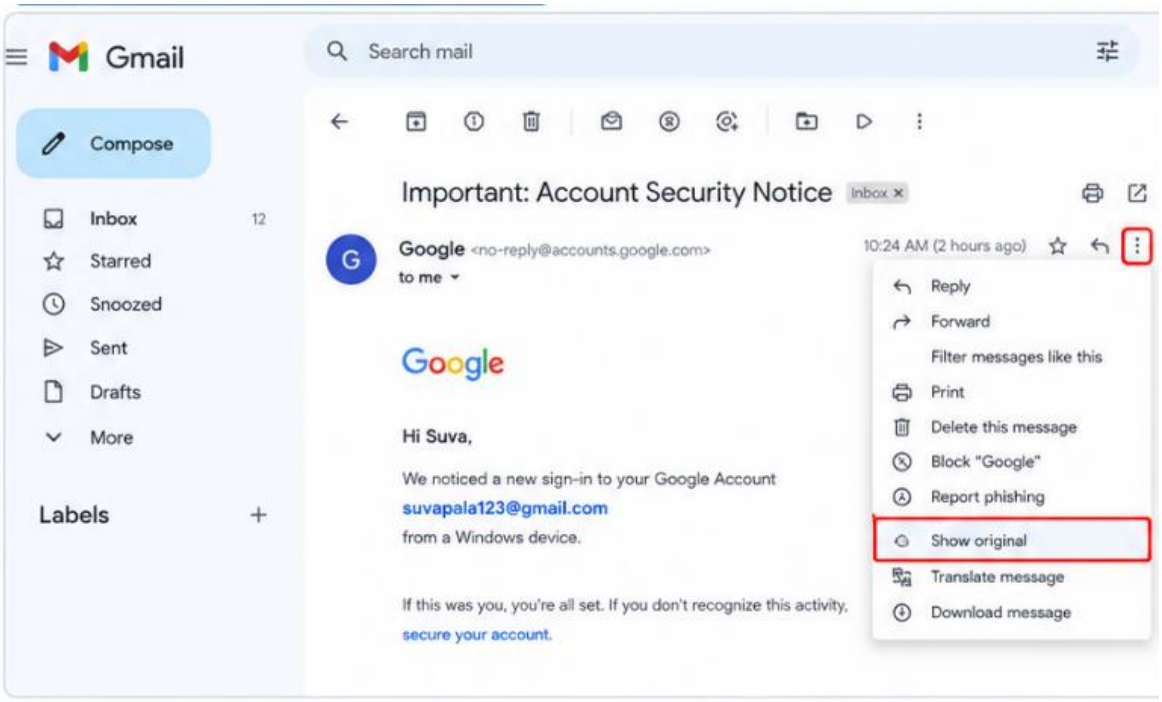
Technique 1: Enable Two-Factor Authentication (2FA) :-

I enabled Two-Factor Authentication (2FA) on my email account. After entering my password, the account requires an additional verification step. This provides an extra layer of security because a person who obtains my password cannot easily access my email account without the second verification factor.



Technique 2: Checking Email Headers

I checked the email header of a received email using Gmail's Show original option. I checked the sender information and authentication results such as SPF and DKIM. These details help identify suspicious or spoofed emails. If an email shows authentication problems or an unexpected sender domain, I will avoid clicking links, downloading attachments, or replying to the email.



5. Explain how a phishing attack could target a Zomato or Swiggy user, and write a step-by-step guide for a new user on how to recognize and avoid such attacks. Include at least 3 warning signs to look for in fake emails or websites.

Ans.:-

Phishing Attack Targeting Zomato or Swiggy Users

A phishing attack can target a Zomato or Swiggy user by sending a fake email, SMS, WhatsApp message, or website link that looks like it comes from the real company. The attacker may claim that the user's food order has a payment problem, the account needs verification, or a refund is waiting. The fake link may then ask the user to enter their login details, card information, UPI details, OTP, or other sensitive information.

Example of a Phishing Attack

A user receives a message:

"Your Swiggy order payment failed. Click here to complete the payment and avoid cancellation."

The link opens a fake Swiggy-looking website. The user enters their mobile number, password, card/UPI information, or OTP. The attacker can then use this information to steal the account or money.

Step-by-Step Guide to Recognize and Avoid Phishing

Step 1: Check the Sender

Check the email address or phone number carefully. Attackers may use an address that looks similar to the genuine company but contains spelling mistakes or an unusual domain.

Step 2: Check the Link

Before clicking a link, move the mouse over it and check the actual website address. Do not enter information on suspicious or misspelled websites.

Step 3: Do Not Share OTP

Never share an OTP, UPI PIN, card PIN, CVV, or password with anyone claiming to be from Zomato or Swiggy.

Step 4: Verify Through the Official App

If you receive a message about an order, refund, payment, or account problem, open the official Zomato/Swiggy app directly instead of clicking the message link.

Step 5: Check for Urgency

Be suspicious of messages saying things like "Act immediately," "Your account will be blocked," or "Pay now to receive your refund." Attackers use urgency to make victims act without thinking.

Step 6: Report and Delete

If the message appears suspicious, do not click its links or attachments. Report it as phishing/spam and delete it.

At Least 3 Warning Signs

1. **Suspicious or misspelled website URL** – The website address is different from the official website.
2. **Urgent or threatening message** – The message pressures the user to act immediately.
3. **Request for sensitive information** – It asks for passwords, OTPs, UPI PINs, CVV, or card details.
4. **Unknown sender** – The email or message comes from an unusual or suspicious address.
5. **Unexpected refund or prize offer** – It promises a refund, discount, coupon, or reward that the user did not request.

Conclusion

A Zomato or Swiggy user can avoid phishing by checking the sender and website URL, avoiding suspicious links, never sharing OTPs or PINs, and using the official app to verify orders and payments. Being careful before clicking or entering information can prevent account theft and financial loss.